DAY 08 ASSIGNMENT

CHALLENGE SUBMISSION — TASKS 1, 4 & 5 (INVESTIGATION EVIDENCE)

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Assignment Code: DAY08-CHALLENGE

Network Identity, DNS & Network Route Investigation.

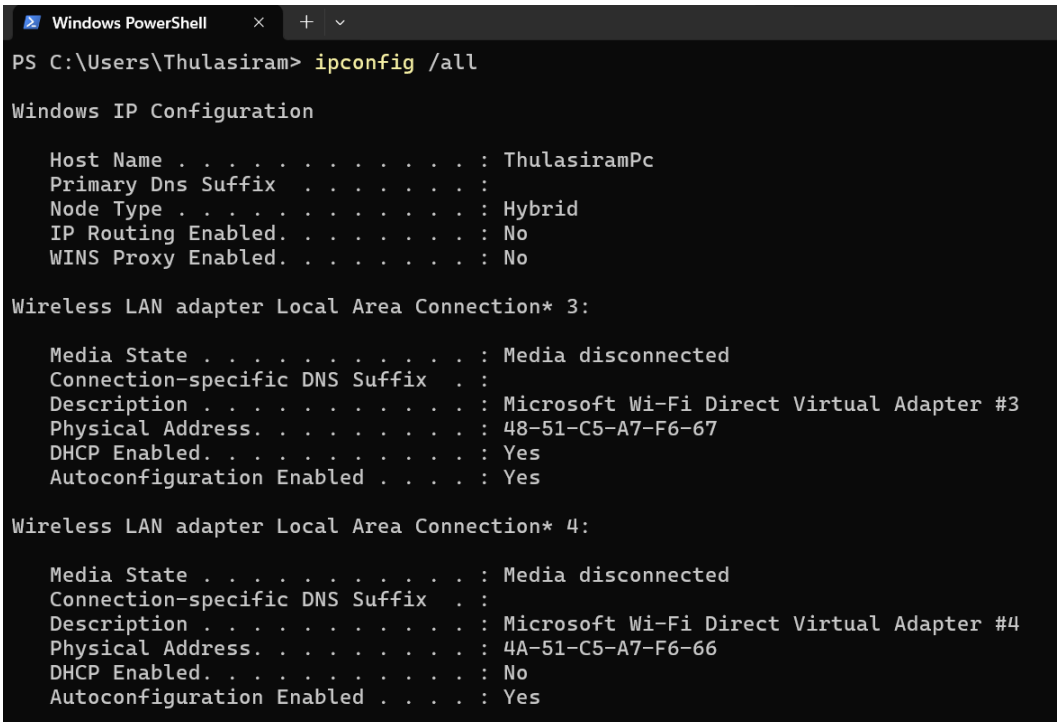
1) TASK 1 - NETWORK IDENTITY

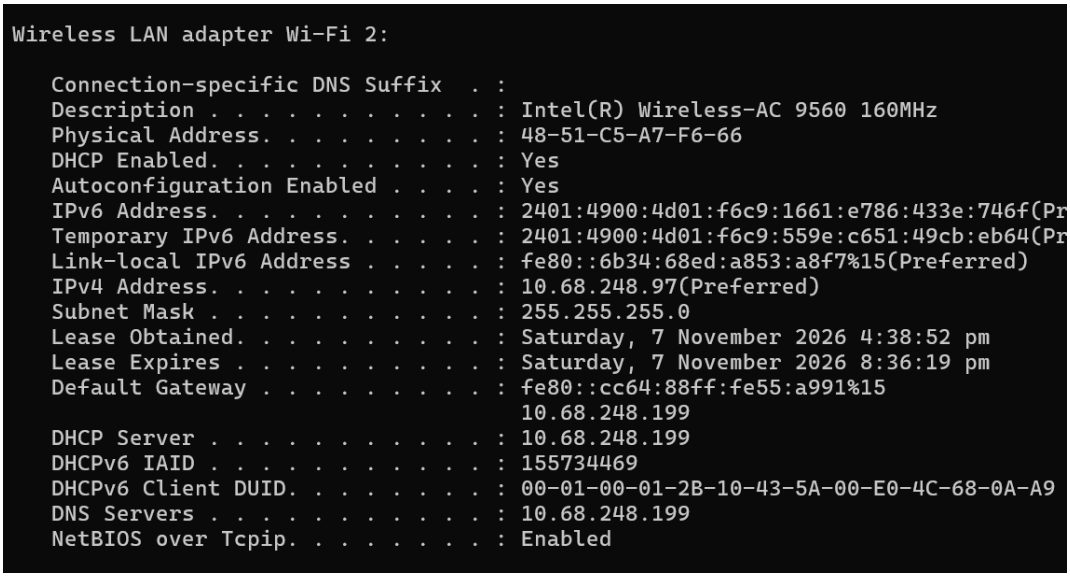
I used THE ipconfig /all command within THE Windows Command Prompt to identify my network configuration. Within my active network adapter I documented THE following:

Network Info	My Network Info
IPv4 Address	10.68.248.97
Subnet Mask	255.255.255.0
Default Gateway	10.68.248.199
DNS Server	10.68.248.199
MAC/Physical Address	48-51-C5-A7-F6-66
DHCP Enabled	yes

Screenshot 1 - Network Configuration

This image shows THE output of THE ipconfig /all command. It shows my network configuration and key information from my active network adapter.





2) TASK 4 - DNS INVESTIGATION

I used THE nslookup google.com command to investigate how THE domain NAME Google.com translates into an IP address.

DNS Server That Responded: 10.68.248.199

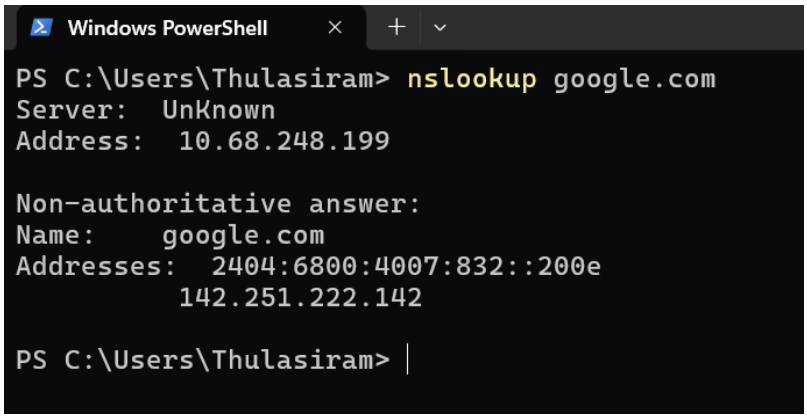
IP Address Returned: 142.251.222.142

What is DNS?

DNS stands for Domain Name System. It is used to translate domain names into IP addresses that can be recognised by computers. This helps make it easier to access websites as you won't have to remember THE numerical IP address.

Screenshot 2 - DNS Investigation

This image shows THE output of THE nslookup google.com command. It shows what DNS SERVER was used and what IP address(ES) were returned.



3) TASK 5 - NETWORK ROUTE INVESTIGATION

I used THE tracert google.com command to investigate THE network route.

Number Of Hops Shown: 5

First Hop: 10.68.248.199

What Does The First Hop Usually Represent?

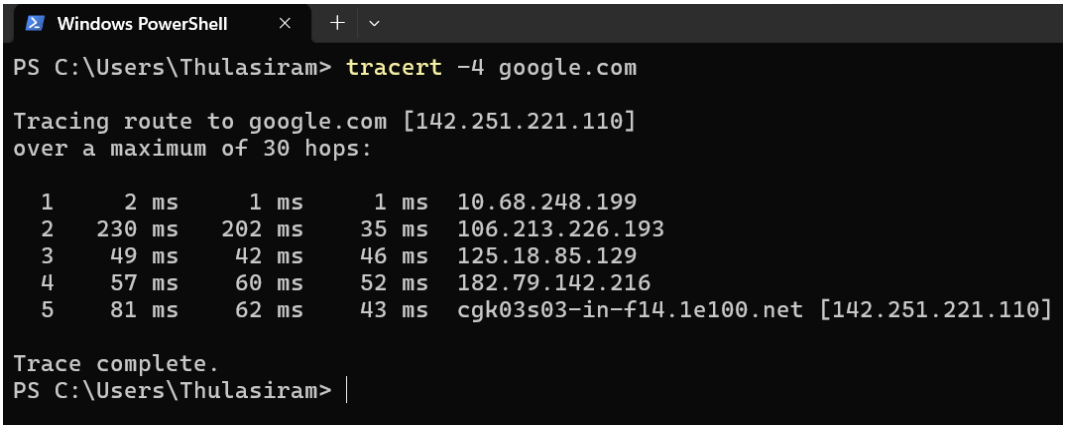
The FIRST HOP will usually represent YOUR local default gateway router.

What is Traceroute Used For?

Traceroute is used to see what route network packets travel between devices. This helps identify what routers are used between devices, understand how THE network is routed and help with any connectivity OR routing issues.

Screenshot 3 - Network Route Investigation

This image shows THE output of THE tracert google.com command. It shows how many network hops there are, THE FIRST HOP and THE route to Google's SERVER.



CONCLUSION

Throughout completing this assignment I have learnt how to identify my network configuration on my computer. I used THE ipconfig /all command to identify my network information, THE nslookup google.com command to understand how DNS works and THE tracert google.com command to see THE network route. This has helped me understand how devices communicate with servers. This knowledge has helped me understand how network professionals investigate connectivity and routing issues. This knowledge can be useful within THE cybersecurity industry to identify network devices, understand how they communicate and investigate any suspicious network activity.